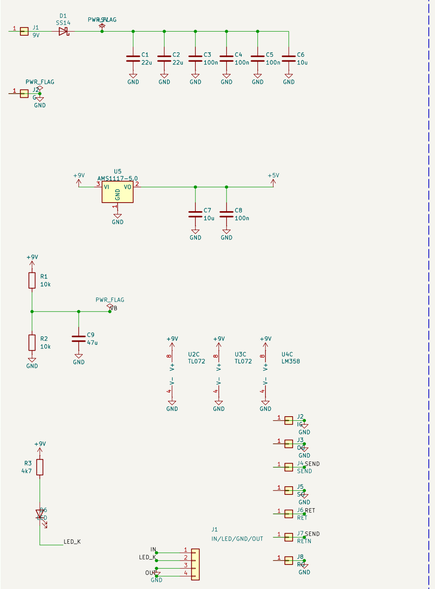
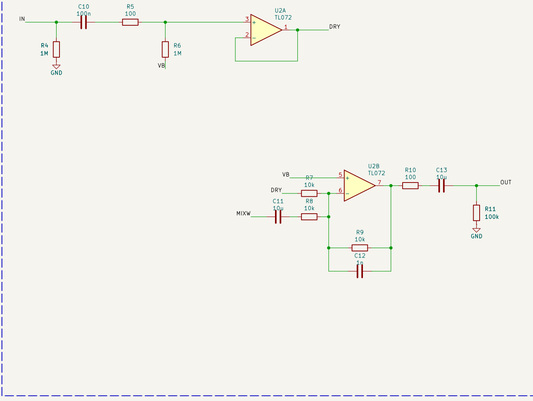
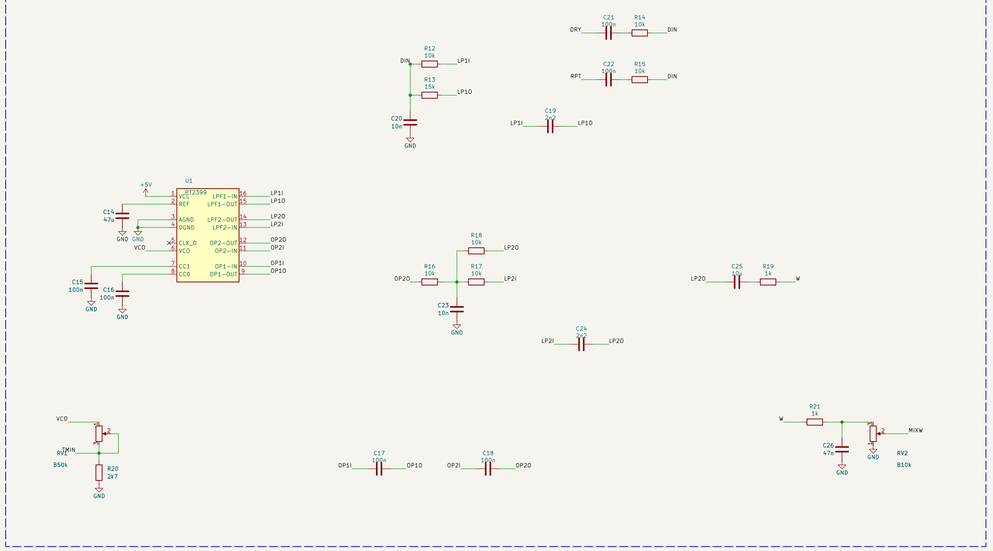
POWER (SS14, AMS1117-5.0 for the PT2399), VB, STATUS LED, FOOTSWITCH HEADER, JACK PADS



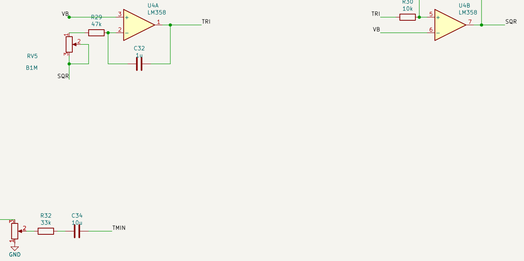
INPUT BUFFER (U2A) + OUTPUT MIXER (U2B): as Small Time



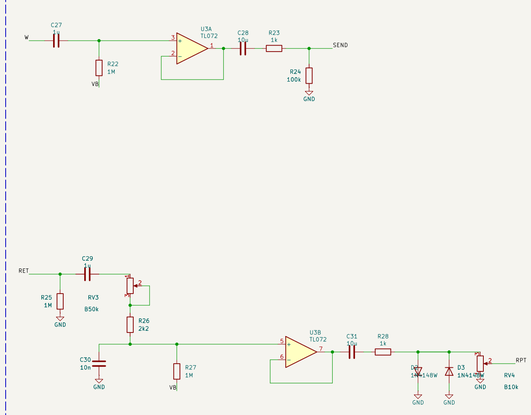
DELAY: PT2399 (U1), Small Time network. TIME = 850k + 2k7 (VCO to ground); the LFO enters at TMIN



MODULATION: LFO (U4, LM358) -> DEPTH -> 33k + 10u -> TMIN (even depth across TIME). ~0.5–12 Hz



FEEDBACK LOOP: SEND (U3A) -> loop -> RETURN (no plug = internal) -> TONE -> U3B -> limiter -> REPEAT



No plug in RETURN: the jack's switch ties RET to SEND (internal path). TONE acts on every repeat in both cases.

A fuzz in the loop adds gain every pass: the 1N4148W pair caps the loop at about +-0.6 V.

PT2399 network from Small Time; SMD for JLC (all basic parts except the PT2399)
PT2399 delay, effects loop in the feedback (SEND / switched RETURN), TONE on every repeat, TIME m

Super-Freq

Sheet: /
File: LoopEcho.kicad_sch

Title: Loop Echo

Size: A1	Date: 2026-10-25	**Rev: 1.0**
KiCad E.D.A. 10.0.4		Id: 1/1